

ATIF KHAN

Embedded Hardware Design Engineer | Power Electronics & PCB Design

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PROFESSIONAL SUMMARY

Embedded Hardware Design Engineer with **4+ years of experience** spanning **schematic design**, multilayer PCB layout, and embedded systems development for **industrial, IoT, wearable**, and consumer electronics **products**. Proven expertise in **power electronics** architecture **DC-DC** conversion, battery management systems (**BMS**), and power-path switching alongside **RF front-end design** and **ATEX/IECEX-compliant** hardware for **hazardous-area** applications. Proficient in **Altium Designer** and **KiCad** for **high speed, impedance-controlled, multilayer** board design, with hands-on experience in MCU **bring-up**, hardware-in-the-loop (**HIL**) validation, and **EMI/EMC** mitigation. Holds a **Bachelor of Science in Engineering Sciences (Optoelectronics)** from **GIKI**. Skilled at translating client and product requirements into reliable, **manufacturable, production ready** hardware from concept through **certification**.

CORE COMPETENCIES

- **PCB Design & Layout:** 2–8 layer, multilayer, impedance-controlled, mixed-signal and RF boards
- **Power Electronics:** Buck/Boost/Buck-Boost converters, LDOs, Li-Ion/LiPo battery management (**BMS**), fuel gauging, power-path switching, reverse-polarity & ESD/TVS protection
- **Embedded Systems:** ARM Cortex-M (STM32, nRF52840), ESP32/ESP32-S3/C6, AVR, 8051, PIC, MCU bring-up and firmware integration
- **Compliance & Standards:** ATEX/IECEX hazardous-area design, IEC 60079-7:2013, QA & compliance verification for Zone 1/Zone 2 deployments
- **RF & Wireless:** Antenna matching, LTE/NB-IoT, BLE 5.0, Wi-Fi 6, GNSS, LORA microstrip impedance-matched RF traces
- **Signal Integrity & EMI/EMC:** Thermal modelling, power integrity optimization, noise mitigation
- **Validation & Testing:** Hardware-in-the-Loop (**HIL**) simulation, oscilloscope/function-generator bench testing, fault isolation
- **Program Delivery:** BOM management, DFM/DFT, component sourcing, cross-functional project management

TECHNICAL SKILLS

- **EDA / CAD Tools:** Altium Designer, KiCad 9.0, EasyEDA Pro, LTspice, Proteus, SolidWorks, FreeCAD
- **Microcontrollers & SoCs:** STM32, ESP32 / ESP32-S3 / ESP32-C6, Nordic nRF52840, AVR (ATtiny, ATmega), AT89S52 (8051), PIC, Renesas Synergy, MAX78000 edge-AI SoC
- **Communication Protocols & Interfaces:** SPI, I2C, UART, CAN, RS-232, RS-485, USB, Ethernet, JTAG/SWD, UPDI
- **Wireless / IoT:** Quectel & SIMCom LTE-M/NB-IoT modules, BLE 5.0, Wi-Fi 6, GNSS, NFC, LORA.
- **Programming & Simulation:** C, C++, Python, Embedded C, MATLAB, Simulink
- **Other:** Microsoft Office, WordPress, HTML/CSS, technical documentation, project management

PROFESSIONAL EXPERIENCE

Embedded Hardware & Power Electronics Design Lahore, Pakistan

Jan 2026 – Present

Embedded Designer (embeddeddesigner.com)

- **Design and develop** embedded hardware and **power management** circuits, covering **schematic capture**, **multilayer** PCB layout, and prototype validation for **client products**.
- Architect power delivery subsystems including **DC-DC** conversion, **battery charging**, and power-path switching for embedded and **IoT**-enabled devices.
- **Select components** and define design-for-manufacturability (**DFM**) requirements in collaboration with cross functional engineering teams.

- Perform hardware **bring-up, debugging, and validation testing** to ensure design reliability and compliance with project specifications.

Hardware Design Engineer *Lahore, Pakistan*

Jan 2025 – Jul 2025

Smart Computings Pvt Ltd (www.smartcomputings.com)

- **Led schematic** verification and **PCB-level** design analysis for an **STM32F407**-based system integrating **4G LTE** connectivity, **relay control**, **SoC** switching between primary and secondary power sources, and **3-phase AC monitoring**.
- Conducted extensive **lab prototyping** and **testing** of digital I/O circuits using **oscilloscopes** and **function generators** for signal verification and fault isolation.
- Validated peripheral interfaces (**SPI, UART, I2C**) and industrial communication protocols including **RS-232, RS-485, CAN, USB, and Ethernet**.
- Performed board bring-up and power-domain verification for a **Renesas Synergy** MCU, ensuring correct voltage sequencing and regulator configuration.
- Integrated **SIMCom** and **Quectel cellular** modules for remote telemetry and validated pressure/temperature sensor interfaces.
- Simulated Hardware-in-the-Loop (**HIL**) conditions and proposed circuit enhancements, including reverse-polarity protection, flyback diode optimization, and **ESD/TVS** placement strategies.
- Reverse-engineered European **ATEX certified** EVC hardware to localize components and reduce cost while preserving safety and functional integrity.
- Designed and tested industrial power regulation modules (24V to 5V/3.3V) in accordance with **IEC 60079-7:2013** for explosive atmospheres.
- Analysed schematic and design compliance for **Zone 1 and Zone 2 ATEX gas-flow** computers, ensuring critical safety and environmental standards were met.
- **Led QA** and compliance verification for field deployment in hazardous (Zone 2) locations, including final assembly checks.
- Managed **firmware flashing**, device initialization, and complete hardware validation of ATEX-compliant systems.
- Used Altium Designer for complex multilayer, impedance-controlled PCB layout, thermal modelling, **EMI mitigation**, and power integrity optimization.
- Oversaw hardware QA, **BOM** verification, and delivery of **production ready** hardware modules for embedded field deployments.

Research & Development (R&D) Engineer *Taxila, Pakistan*

Jun 2024 – Dec 2024

NR Trading Pvt Ltd. *Taxila, Pakistan*

- Designed **multilayer** and **RF PCBs**, handling **component selection** and design optimization.
- Developed embedded systems using **AT89S52, STM32, and PIC** microcontrollers, including **schematic** design and **simulation**.
- **Reverse-engineered** existing PCBs and delivered enhanced **redesigns** with improved performance.
- Managed **prototyping, testing, and troubleshooting** to ensure high-quality, on-time project delivery.

Embedded Systems Firmware Developer *Peshawar, Pakistan*

Aug 2023 – Sep 2023

Chip Soul Technology Pvt Ltd

- Gained **hands-on experience in embedded systems** development and C programming for the **8051-microcontroller** family.
- Worked on microcontroller interfacing and control-system projects.

Hardware Design Engineer *Peshawar, Pakistan*

Jul 2022 – Sep 2022

Chip Soul Technology Pvt Ltd

- Learned detailed **PCB** design practices, including **multilayer** board development.
- Gained experience in **component selection** for optimized circuit performance.
- Assisted in **layout** and **simulation** of electronic circuits across **multiple projects**.

Embedded Hardware Design Engineer

Dec 2021 – Present

Remote / Online

- Deliver end-to-end **embedded** hardware solutions for **global** clients, spanning schematic design, **multilayer PCB** layout, component selection, and hardware **bring-up** for embedded, IoT, and **power electronics** systems.

- **Collaborate with clients** from concept through **prototyping** and **validation**, ensuring designs meet **performance**, reliability, and **manufacturability** requirements in line with industry best practices.

KEY PROJECTS

AI-Enabled Electrical Fault Detector — Edge-AI Sensor Node

- Designed a 6-layer, hierarchical mixed-signal PCB (KiCad 9.0) for an edge-AI arc-fault detection sensor node powered by the MAX78000 (Arm Cortex-M4F + CNN hardware accelerator) for on-device, real-time fault classification.
- Built an arc-fault analog front-end based on TI's TIDA-01385/TIDA-010231 reference design with CT sensing, active filtering, and a 12-bit SAR ADC; implemented CT-based energy harvesting and a 4-rail (1.2V/1.8V/2.5V/3.3V) power architecture.
- Enabled dual-radio IoT connectivity (LTE-M/NB-IoT + Wi-Fi 6/BLE) with multi-modal sensor fusion (IMU, environmental, acoustic) and full Li-Ion battery management with fuel gauging.

BLE Motion & Biometric Training Device — Wearable Sensor

- Designed a compact 4-layer wearable PCB (Altium Designer) around the Nordic nRF52840 (BLE 5 + NFC tap-to-pair) with a 6-axis IMU for motion, posture, and cadence tracking.
- Engineered complete power architecture: Li-Po charging with Model Gauge fuel gauging, buck-boost regulation, and a pushbutton power controller for clean on/off and kill-switch operation.
- Implemented USB-C charging/programming, capacitive touch HMI, and on-board NAND flash logging for offline-capable training sessions.

Smart Liquid Dispensing & Motor Control Device (KIPFLO)

- Designed a 4-layer PCB (KiCad 9.0) built around the ESP32-S3, integrating a dual H-bridge motor driver, Hall-effect position sensing, capacitive liquid-level sensing, and a 3-axis accelerometer for orientation/fall detection.
- Implemented a complete battery charging system with Dynamic Power Path Management (DPPM) and a POGO-pin charging dock with automatic power-path switching for safe charging.

Multi-Rail Power Distribution Unit (12V, 9A)

- Designed a 2-layer power distribution PCB (Altium Designer) supporting three independently fused 12V output rails (9A/3A/5A) for industrial and automotive applications.
- Implemented ideal-diode reverse-polarity protection, dual TVS surge protection, and high-current vibration-tolerant connectors for robust field deployment.
- Additional projects — including a cellular IoT GNSS tracker, an AI-powered OCR text-recognition device, and a low-power wearable digital counter — are detailed in the [GitHub portfolio](#).

EDUCATION

B.S. in Engineering Sciences — Optoelectronics

Sep 2020 – Jun 2024

Ghulam Ishaq Khan Institute of Engineering Sciences & Technology (GIKI), Swabi, Pakistan

- Final Year Project: NOMA-based Visible Light Communication (VLC) system for two users covering modulation, user separation, interference analysis, and system performance evaluation.